Sheth L.U.J. And Sir M.V. College

1.A Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques

INPUT:-

```python
def encrypt(text, shift):
    result = ""

    for char in text:
        if char.isalpha():
            ascii_offset = 65 if char.isupper() else 97
            result += chr((ord(char) - ascii_offset + shift) % 26 + ascii_offset)
        else:
            result += char

    return result


def decrypt(text, shift):
    return encrypt(text, -shift)


while True:
    print("\n===== Caesar Cipher =====")
    print("1. Encrypt")
    print("2. Decrypt")
    print("3. Exit")

    choice = input("Enter Choice: ")
```

```python
    print("\n===== Caesar Cipher =====")
    print("1. Encrypt")
    print("2. Decrypt")
    print("3. Exit")

    choice = input("Enter Choice: ")

    if choice == "1":
        msg = input("Enter Message: ")
        key = int(input("Enter Shift Key: "))
        print("Encrypted Message:", encrypt(msg, key))

    elif choice == "2":
        msg = input("Enter Cipher Text: ")
        key = int(input("Enter Shift Key: "))
        print("Decrypted Message:", decrypt(msg, key))

    elif choice == "3":
        print("Thank You")
        break

    else:
        print("Invalid Choice")
```

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Roll No. T093

Subject:- Cyber And Information Security

OUTPUT:-

```
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C:\Users\CHETAN\Documents>"C:\Users\CHETAN\Documents\Cyber Security Practical No. 1A.py"

===== Caesar Cipher =====
1. Encrypt
2. Decrypt
3. Exit
Enter Choice: 1
Enter Message: Hello From TYCS
Enter Shift Key: 5
Encrypted Message: Mjqqt Kwtr YDHX

===== Caesar Cipher =====
1. Encrypt
2. Decrypt
3. Exit
Enter Choice: 2
Enter Cipher Text: Hello World
Enter Shift Key: 4
Decrypted Message: Dahhk Sknhz

===== Caesar Cipher =====
1. Encrypt
2. Decrypt
3. Exit
Enter Choice: 3
Thank You

C:\Users\CHETAN\Documents>
```

GUI VERSION:-

Cyber Security Practical No. 1A.py - C:/Users/CHETAN/Documents/Cyber Security Practical No. 1A.py (3.12.4)

```python
from tkinter import *

def encrypt():
    text = message.get()
    shift = int(key.get())
    result = ""

    for ch in text:
        if ch.isalpha():
            offset = 65 if ch.isupper() else 97
            result += chr((ord(ch)-offset+shift)%26+offset)
        else:
            result += ch

    output.delete(0, END)
    output.insert(0, result)


def decrypt():
    text = message.get()
    shift = int(key.get())
    result = ""
```

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Cyber Security Practical No. 1A.py - C:/Users/CHETAN/Documents/Cyber Security Practical No. 1A.py (3.12.4)

```
def decrypt():
    text = message.get()
    shift = int(key.get())
    result = ""

    for ch in text:
        if ch.isalpha():
            offset = 65 if ch.isupper() else 97
            result += chr((ord(ch)-offset-shift)%26+offset)
        else:
            result += ch

    output.delete(0, END)
    output.insert(0, result)


root = Tk()
root.title("Caesar Cipher")
root.geometry("400x300")

Label(root, text="Message").pack()

message = Entry(root, width=40)
```

Cyber Security Practical No. 1A.py - C:/Users/CHETAN/Documents/Cyber Security Practical No. 1A.py (3.12.4)

```
root = Tk()
root.title("Caesar Cipher")
root.geometry("400x300")

Label(root, text="Message").pack()

message = Entry(root, width=40)
message.pack()

Label(root, text="Shift Key").pack()

key = Entry(root)
key.pack()

Button(root, text="Encrypt", command=encrypt).pack(pady=5)
Button(root, text="Decrypt", command=decrypt).pack(pady=5)

output = Entry(root, width=40)
output.pack()

root.mainloop()
```

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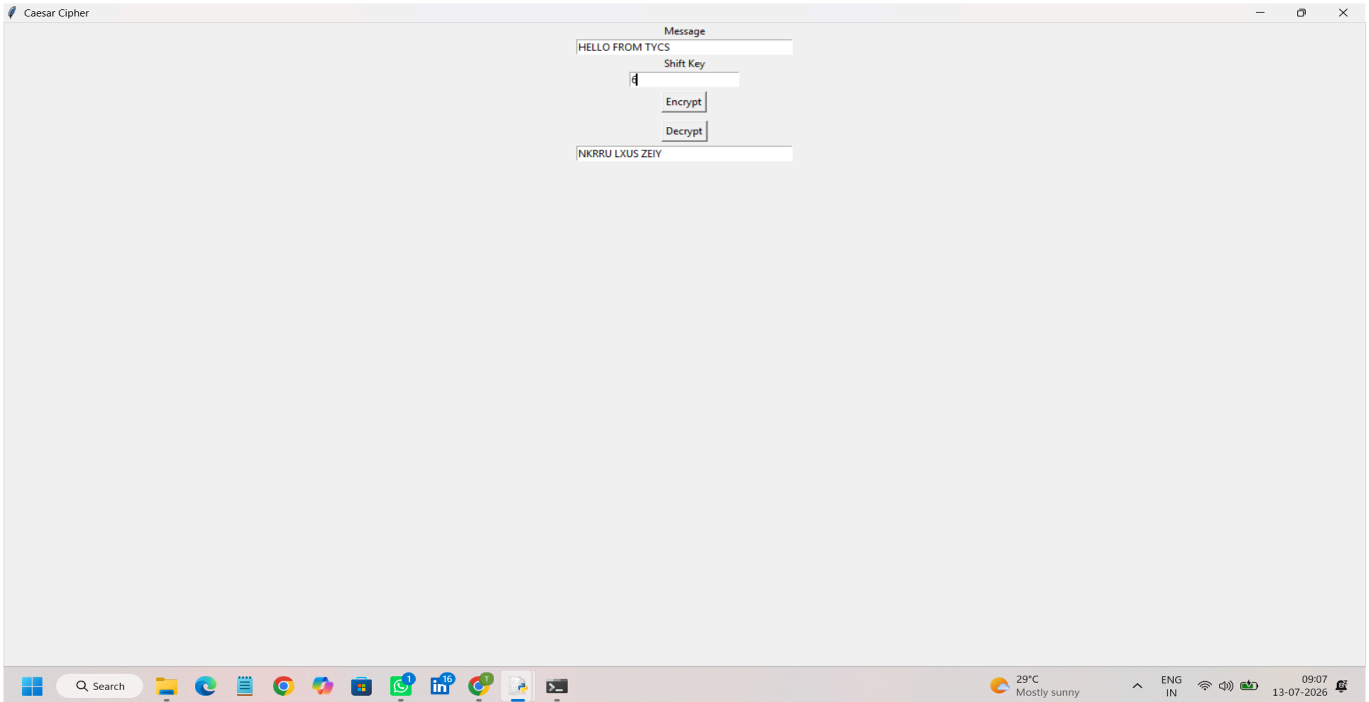
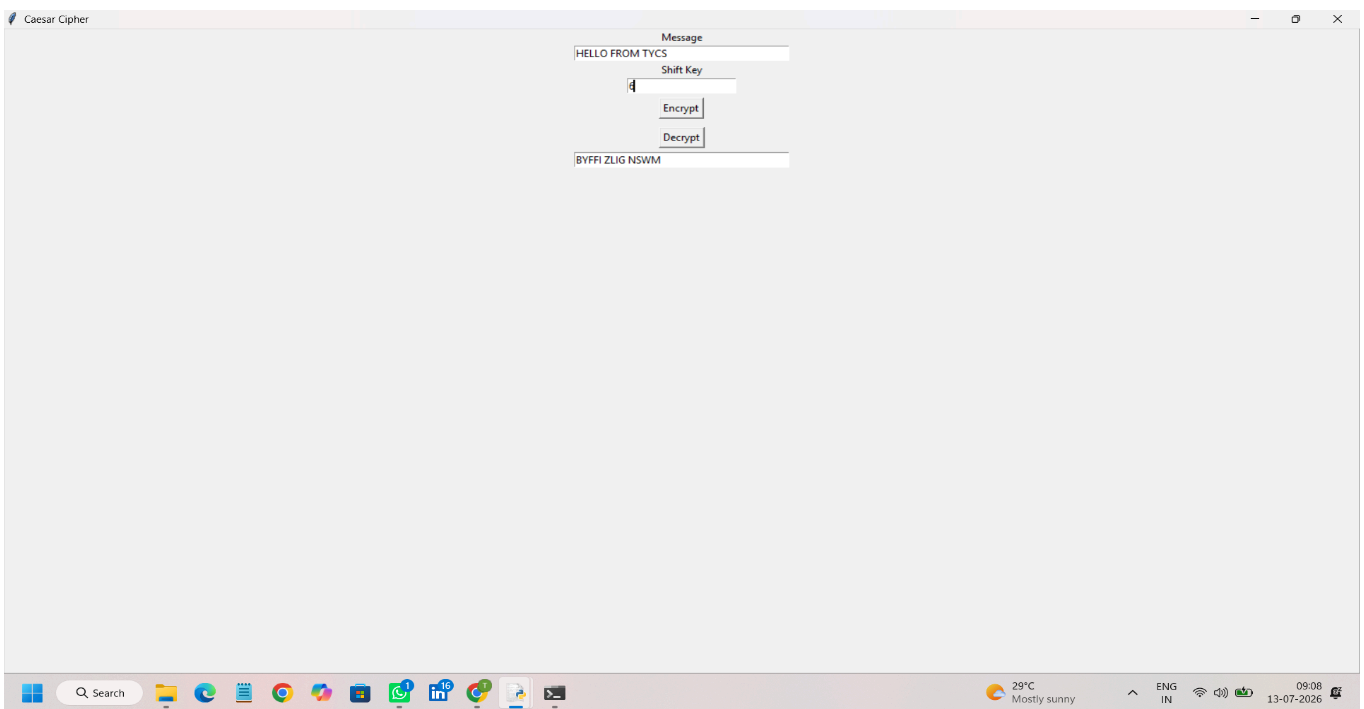
OUTPUT:-


Caesar Cipher
Message
HELLO FROM TYCS
Shift Key
6
Encrypt
Decrypt
NKRRU LXUS ZEIY
Search
29°C
Mostly sunny
ENG
IN
09:07
13-07-2026


Caesar Cipher
Message
HELLO FROM TYCS
Shift Key
6
Encrypt
Decrypt
BYFFI ZLIG NSWM
Search
29°C
Mostly sunny
ENG
IN
09:08
13-07-2026

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1. B Design and implement algorithms to encrypt and decrypt messages using transposition techniques

INPUT:-

Cyber Security Practical No. 1B.py - C:/Users/CHETAN/Documents/Cyber Security Practical No. 1B.py (3.12.4)

```python
def encrypt(text):
    even = text[::2]
    odd = text[1::2]
    return even + odd


def decrypt(cipher):
    half = (len(cipher)+1)//2
    even = cipher[:half]
    odd = cipher[half:]

    result = ""

    for i in range(half):
        result += even[i]
        if i < len(odd):
            result += odd[i]

    return result


while True:

    print("\n===== Rail Fence Cipher =====")
```

Cyber Security Practical No. 1B.py - C:/Users/CHETAN/Documents/Cyber Security Practical No. 1B.py (3.12.4)

```python
while True:

    print("\n===== Rail Fence Cipher =====")
    print("1. Encrypt")
    print("2. Decrypt")
    print("3. Exit")

    choice = input("Enter Choice: ")

    if choice == "1":
        msg = input("Enter Message: ")
        print("Encrypted:", encrypt(msg))

    elif choice == "2":
        msg = input("Enter Cipher Text: ")
        print("Decrypted:", decrypt(msg))

    elif choice == "3":
        break

    else:
        print("Invalid Choice")
```

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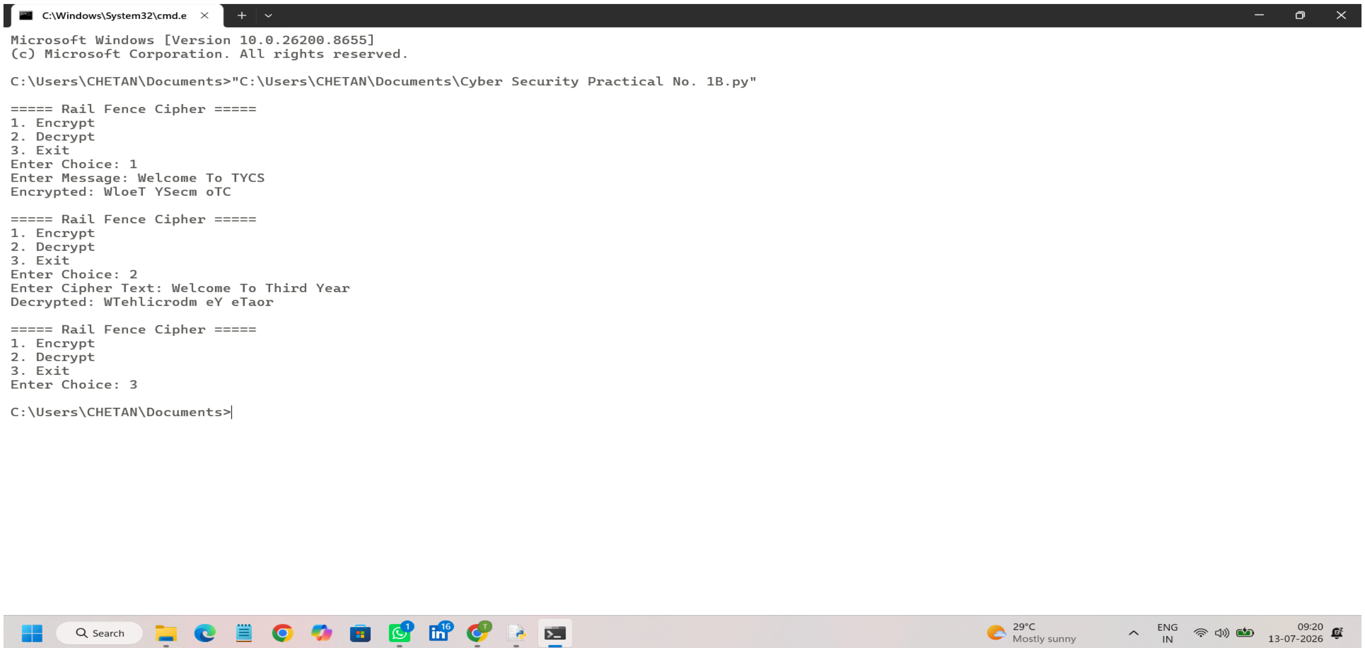
OUTPUT:-



```
Microsoft Windows [Version 10.0.26200.8655]
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C:\Users\CHETAN\Documents>"C:\Users\CHETAN\Documents\Cyber Security Practical No. 1B.py"

===== Rail Fence Cipher =====
1. Encrypt
2. Decrypt
3. Exit
Enter Choice: 1
Enter Message: Welcome To TYCS
Encrypted: WloeT YSecm oTC

===== Rail Fence Cipher =====
1. Encrypt
2. Decrypt
3. Exit
Enter Choice: 2
Enter Cipher Text: Welcome To Third Year
Decrypted: WTehlicrodm eY eTaor

===== Rail Fence Cipher =====
1. Encrypt
2. Decrypt
3. Exit
Enter Choice: 3

C:\Users\CHETAN\Documents>
```
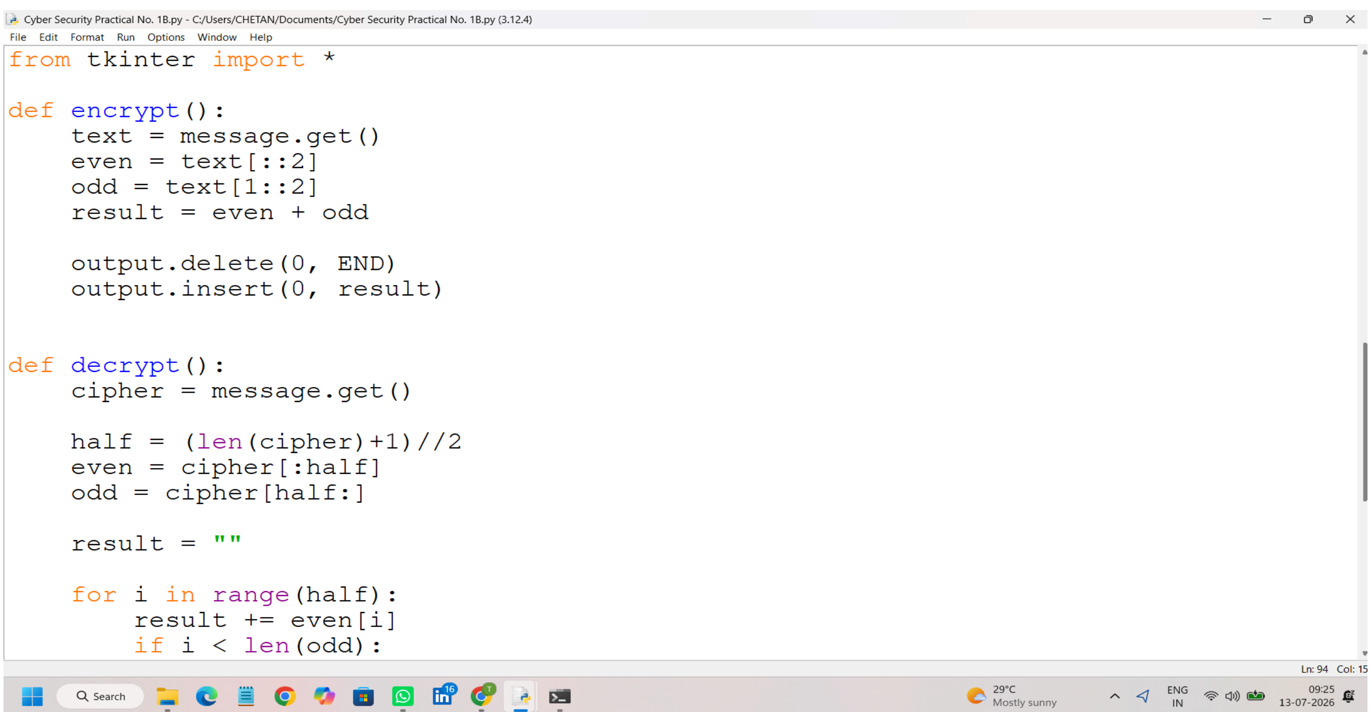

GUI VERSION:-



```python
from tkinter import *

def encrypt():
    text = message.get()
    even = text[::2]
    odd = text[1::2]
    result = even + odd

    output.delete(0, END)
    output.insert(0, result)


def decrypt():
    cipher = message.get()

    half = (len(cipher)+1)//2
    even = cipher[:half]
    odd = cipher[half:]

    result = ""

    for i in range(half):
        result += even[i]
        if i < len(odd):
```

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Cyber Security Practical No. 1B.py - C:/Users/CHETAN/Documents/Cyber Security Practical No. 1B.py (3.12.4)

```python
    for i in range(half):
        result += even[i]
        if i < len(odd):
            result += odd[i]

    output.delete(0, END)
    output.insert(0, result)


root = Tk()

root.title("Rail Fence Cipher")
root.geometry("400x250")

Label(root, text="Message").pack()

message = Entry(root, width=40)
message.pack()

Button(root, text="Encrypt", command=encrypt).pack(pady=5)
Button(root, text="Decrypt", command=decrypt).pack(pady=5)

output = Entry(root, width=40)
output.pack()
```

Cyber Security Practical No. 1B.py - C:/Users/CHETAN/Documents/Cyber Security Practical No. 1B.py (3.12.4)

```python
            result += odd[i]

    output.delete(0, END)
    output.insert(0, result)


root = Tk()

root.title("Rail Fence Cipher")
root.geometry("400x250")

Label(root, text="Message").pack()

message = Entry(root, width=40)
message.pack()

Button(root, text="Encrypt", command=encrypt).pack(pady=5)
Button(root, text="Decrypt", command=decrypt).pack(pady=5)

output = Entry(root, width=40)
output.pack()

root.mainloop()
```

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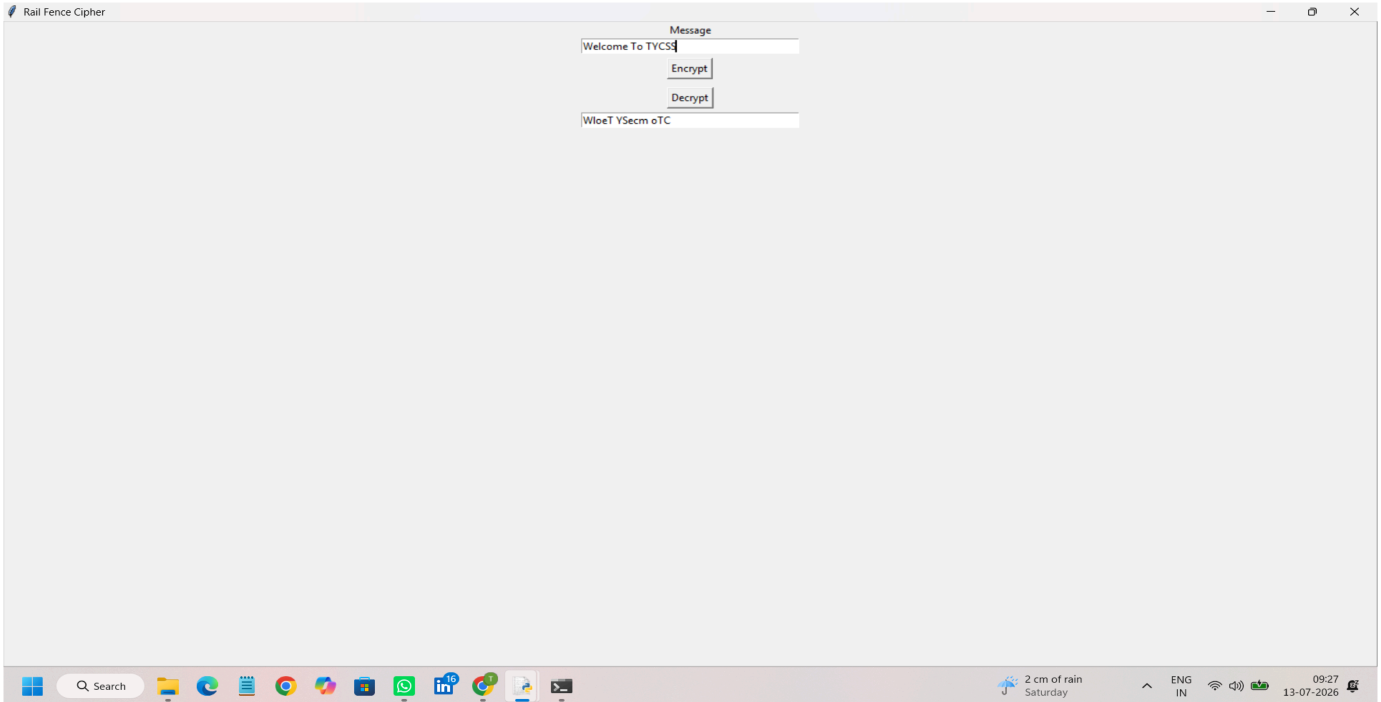
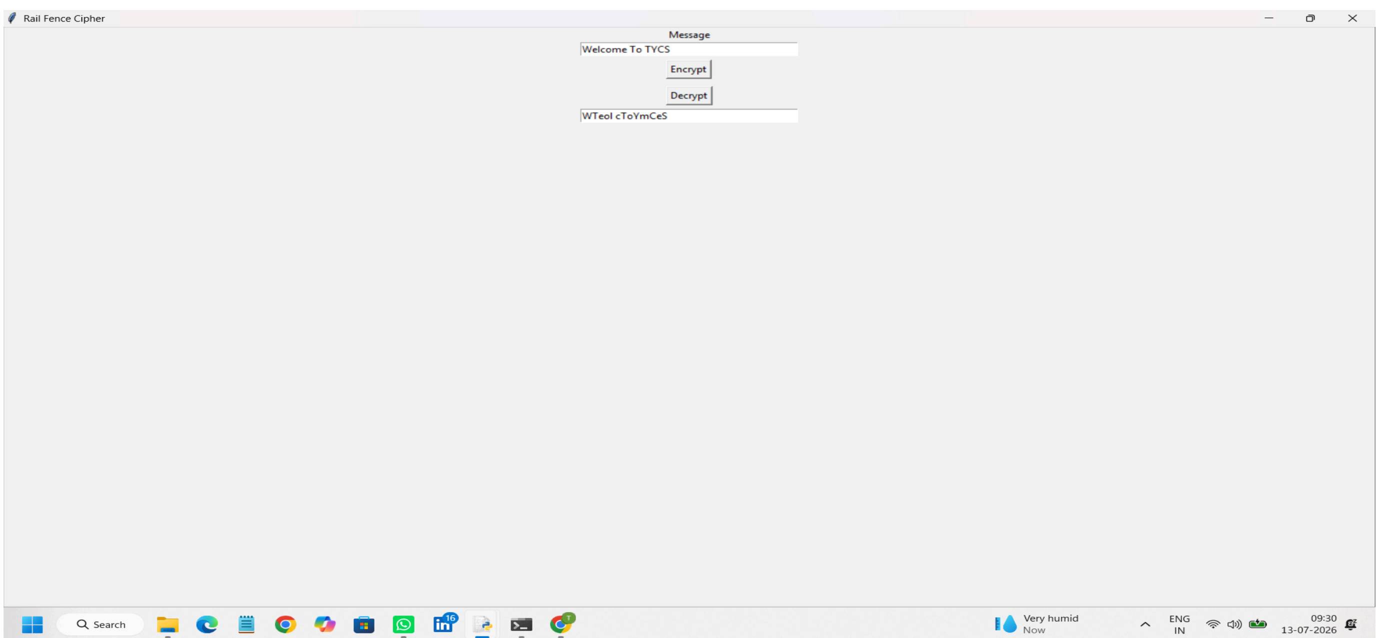
OUTPUT:-





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